

AI-POWERED MENTAL HEALTH THERAPY PLATFORM WITH EMOTIONAL INTELLIGENCE AND CRISIS DETECTION

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CHAPTER 01 – INTRODUCTION

1.1 Chapter Overview

This chapter introduces the research undertaken in this thesis, titled “AI-Powered Mental Health Therapy Platform with Emotional Intelligence and Crisis Detection.” It presents the background to the problem, the problem statement, the research question, the motivation behind the study, the research aim and objectives, a rich picture of the proposed solution, the resource requirements, and the scope of the project. The chapter concludes with a summary that links these elements to the remaining chapters of the thesis.

1.2 Problem Background

Mental well-being has become one of the major issues in public healthcare. Disorders such as anxiety, depression, stress, and emotional exhaustion continue to affect millions of people regardless of age, gender, and occupation. As reported by the World Health Organization (WHO), approximately one in eight people worldwide live with a diagnosed mental disorder, and university students and young adults are particularly vulnerable because of academic pressure, financial difficulties, social challenges, and other contributing factors [1].

Although mental health problems are increasingly widespread, access to appropriate professional support remains limited for many individuals. Social stigma, treatment costs, limited availability of mental health professionals, and accessibility barriers prevent people from seeking timely assistance. These challenges have increased interest in digital solutions that can provide accessible and continuous emotional support.

Recent advances in Artificial Intelligence (AI), Natural Language Processing (NLP), Speech Processing, and Large Language Models (LLMs) have enabled intelligent conversational systems that can understand natural language and provide emotionally supportive interactions. Modern AI-based systems are able to simulate empathic conversations, recognise emotional triggers in user messages, provide CBT-based advice, suggest mindfulness activities, and track a user’s emotional condition over time [2], [3].

Even though many AI-powered mental health applications already exist, most current solutions focus primarily on text-based conversation. Emotional cues expressed through a person’s voice — tone, pitch, pace, and energy — are frequently ignored, despite carrying valuable information about a person’s psychological state. Combining Speech Emotion Recognition (SER) with text-based emotion detection therefore makes it possible to obtain a more complete

picture of a user’s emotional condition [4], [9]. In addition, most existing AI therapy systems have insufficient capability for recognising users experiencing serious psychological distress or suicidal tendencies [5], [8].

A key development in this research is the move away from relying on a single general-purpose conversational API for emotion classification. Instead, a dedicated AI model is trained to detect the user’s emotional state, and the detected emotional information is then used as an input to a separate crisis detection process. In parallel, the Gemini API is retained, but repositioned: it is customised to act as a supportive, mental-health-aware voice conversational agent rather than as the primary emotion classifier. This research therefore proposes the development of an AI-powered mental health therapy platform that combines a trained emotion and crisis detection model, a customised Gemini-based voice agent, personalised therapeutic recommendations, and mental health analytics, within a single secure, privacy-conscious platform.

1.3 Problem Statement

The number of mental health disorders is growing worldwide despite the inadequate provision of professional help to those affected. Many existing digital mental health applications provide support mainly through text-based conversation, relying on general-purpose conversational AI for both dialogue generation and emotional understanding. While such systems widen access to emotional support, using a general conversational API alone for emotion and crisis identification does not provide the specialised, purpose-built classification that this research problem requires.

1.3.1 General Problem

At a high level, the inadequate integration of emotion recognition, crisis detection, and personalised support in digital mental health tools means that users in genuine psychological distress may not be identified early enough for timely intervention. Because emotional states can be expressed through multiple channels — written text carries semantic information, while voice carries additional expressive and conversational cues — systems that rely on only one modality obtain an incomplete view of the user’s condition. This has broader public-health implications: opportunities for early, low-cost, digitally-mediated intervention are being missed at a time when demand for mental health support is rising faster than the supply of qualified professionals [1], [6].

1.3.2 Specific Problem

Within the AI-for-mental-health domain specifically, three shortcomings stand out. First, emotion detection in most existing platforms depends on a general-purpose conversational API rather than a model trained specifically for the target emotional categories, which limits classification reliability and interpretability. Second, crisis detection is rarely driven directly by structured emotional-state output; where it exists, it is often based on static keyword matching or superficial sentiment scoring, which can fail to recognise complex or gradually escalating patterns of risk [5], [8]. Third, personalisation, mental health analytics, and ethical data handling are inconsistently implemented, so that even where emotional support is provided, it is not adapted to the individual’s emotional history, nor tracked over time in a way that protects user privacy [7], [10].

These shortcomings collectively point to a clear research gap: the absence of an integrated platform in which a dedicated, trained model performs emotion detection, that emotional output is used explicitly to drive crisis detection, a customised conversational voice agent handles supportive dialogue, and personalised recommendations and analytics are layered on top — all within an ethically governed system. Addressing this gap is the focus of the current research.

1.4 Research Question

How can a trained emotion and crisis detection model, together with a customised Gemini API-based voice agent, be used to improve AI-enabled mental health support for users, while maintaining privacy and ethical safeguards?

This main research question can be further explored through the following sub-questions:

- RQ1. How effectively can a trained AI model recognise relevant emotional states from user interactions?
- RQ2. How can detected emotional states be used to identify potential mental health crisis conditions?
- RQ3. How can a customised Gemini API-based voice agent provide context-aware and supportive mental health conversations?
- RQ4. How can personalised therapeutic recommendations based on detected emotions improve user engagement with the platform?
- RQ5. How can users’ mental health information be stored and analysed securely while adhering to ethical AI principles?

1.5 Research Motivation

This research is motivated by the growing gap between the demand for mental health support and the limited availability of affordable, accessible, professional care, particularly among university students and young adults who face significant academic and social pressures. The researcher observed that although numerous AI-powered mental health chatbots already exist (e.g., Woebot, Wysa, and Tess), most treat emotion recognition as a secondary function of a general conversational model, rather than as a dedicated, purpose-trained capability, and few combine text and voice modalities with a structured crisis-detection pathway.

The motivation is therefore twofold: academically, to investigate whether separating emotion/crisis classification (via a trained model) from conversational generation (via a customised voice agent) produces a more reliable and interpretable mental health support system; and practically, to build a working platform that can offer stigma-free, continuously available, and privacy-respecting emotional support, while responsibly directing users toward professional or emergency help when a potential crisis is detected.

1.6 Research Aim

To design and implement an AI-powered mental health therapy platform that uses a trained emotion and crisis detection model together with a customised Gemini API-based voice agent to provide context-aware emotional support, personalised therapeutic guidance, and mental health analytics, while maintaining user privacy and ethical safeguards.

1.7 Research Objectives

Primary Objective

To design and develop an AI-driven mental health support platform capable of emotion detection, crisis detection, supportive voice interaction, personalised therapy recommendations, and mental health analytics.

Specific Objectives

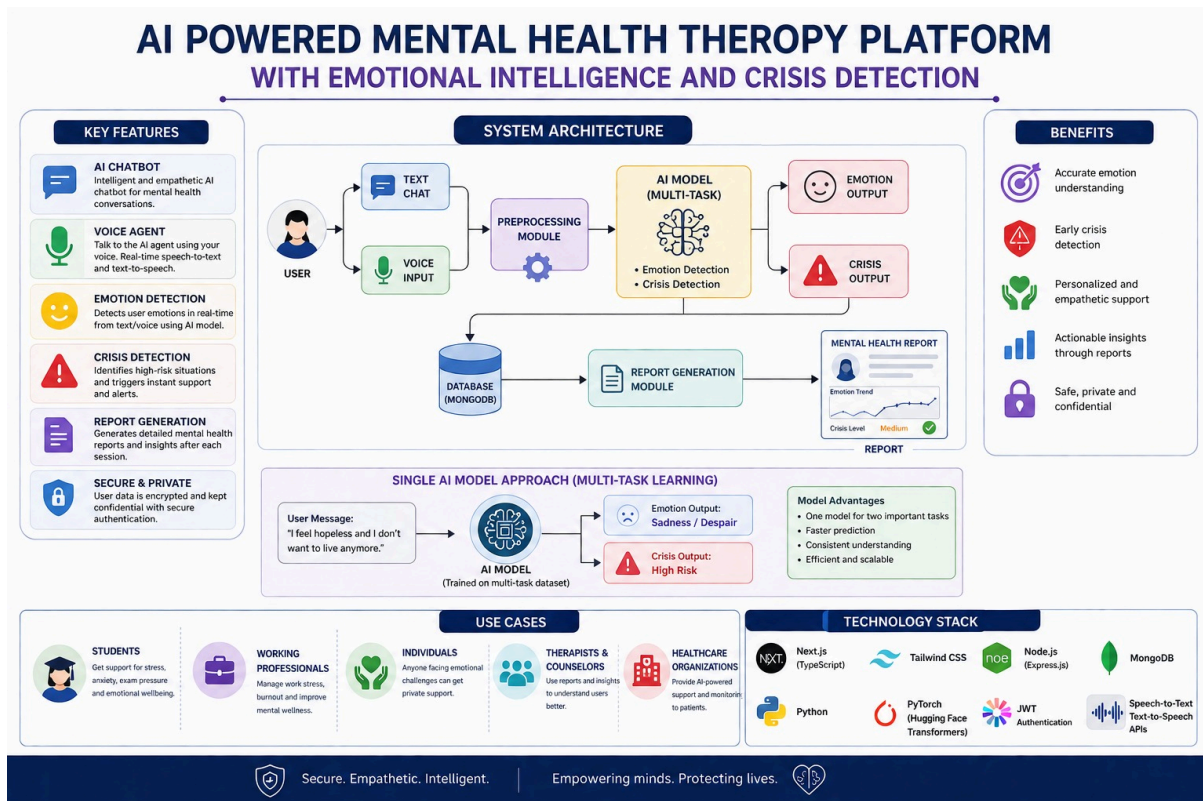
- To identify the functional and non-functional requirements for an integrated emotion-aware, crisis-sensitive mental health platform through a review of existing literature and systems.
- To analyse and train an AI model capable of detecting relevant emotional states from user text and voice interactions.

- To design and develop a crisis detection mechanism that uses the detected emotional states as a primary input for identifying users who may require additional support.
- To design and implement a customised Gemini API-based voice agent that provides natural, context-aware, and supportive mental health conversations.
- To design and develop personalised therapy recommendations (mindfulness activities, breathing exercises, journaling support, and CBT-oriented suggestions) and a secure analytics dashboard for tracking emotional trends.
- To evaluate the developed platform in terms of emotion/crisis model performance, functionality, usability, and user satisfaction.

1.8 Rich Picture of the Proposed Solution

The realised solution, named MindCare, can be visualised as a workflow in which a user interacts with the platform through a therapy chat or voice session. Each user message is first sent to a dedicated Python FastAPI ML microservice, where two trained scikit-learn models classify the input: a TF-IDF + LinearSVC model predicts one of six emotion categories (Sad, Anxious, Angry, Hopeful, Calm, Neutral) with a confidence score, and a TF-IDF + LogisticRegression model predicts one of seven mental health status categories (Anxiety, Bipolar, Depression, Normal, Personality Disorder, Stress, Suicidal). A keyword-based safety net cross-checks the raw text against a curated list of self-harm/suicide-related terms, and the combined result is reduced to a single crisis severity level (none, low, medium, or high).

This crisis level determines the next step deterministically: if the level is medium or high, the Google Gemini LLM is bypassed entirely and a fixed, human-reviewed safety response is returned together with a high-visibility warning banner and the 988 Suicide & Crisis Lifeline number. If the level is none or low, the detected emotion, confidence, and message content are passed to the Gemini API (gemini-3.5-flash-lite), which generates a supportive, structured counselling reply using one of six named techniques (Reflective Listening, Validation, Cognitive Reframing, Grounding, Socratic Questioning, or Practical Coping Suggestion), returned as strict JSON. For voice sessions, user speech is first transcribed to text via Gemini STT, processed through the same pipeline, and the reply is synthesised back to speech via Gemini TTS (voice “Kore”). Every message, its detected emotion, confidence, crisis level, and counselling technique are persisted to MongoDB, alongside separate mood-entry and activity logs, and surfaced back to the user through the dashboard, reports, and history views.



1.9 Resource Requirements

1.9.1 Hardware

- Development workstation/laptop with a minimum of an Intel Core i5 (or equivalent) processor, 16GB RAM, and SSD storage, sufficient for running the Next.js/Node.js development environment and local testing.
- A GPU-enabled environment (either a local GPU or a cloud-based service such as Google Colab or Kaggle Notebooks) for training and validating the emotion/crisis detection model.
- A stable broadband internet connection for accessing the Gemini API, cloud deployment services, and version-control repositories.
- A microphone-equipped device for testing voice-based interaction with the Gemini voice agent.

1.9.2 Software

Component	Technology / Tool
Frontend Framework	Next.js 16 (App Router), React 19
Programming Language	TypeScript
UI Framework	Tailwind CSS v4, Shadcn UI, Radix UI, Framer Motion
Database & ORM	MongoDB, Mongoose
Authentication	Custom JWT sessions (jsonwebtoken, bcryptjs)

ML Microservice	Python 3.10+, FastAPI, Uvicorn, scikit-learn, NumPy, Pydantic
Emotion & Mental-Health Models	TF-IDF + LinearSVC (emotion); TF-IDF + LogisticRegression (mental health status)
Conversational LLM / STT / TTS	Google Gemini API (gemini-3.5-flash-lite; gemini-2.5-flash-preview-tts)
Transactional Email	Resend API
Schema Validation	Zod
Version Control	Git & GitHub
IDE	Visual Studio Code

1.10 Project Scope

The present study focuses on designing and developing an AI-driven online mental health support platform intended primarily for university students and young adults. The platform combines a trained emotion detection model, crisis detection based on the detected emotional states, a customised Gemini API-based voice agent, personalised wellness recommendations, and mental health analytics.

In Scope	Out of Scope
AI-supported text and voice conversation via the customised Gemini voice agent	Replacing licensed psychologists, psychiatrists, or counsellors
Trained model for text/voice emotion detection	Clinical diagnosis of mental health disorders
Emotion-informed crisis detection and escalation guidance	Direct emergency response dispatch (the system encourages, but does not itself contact, emergency services)
Personalised recommendations: mindfulness, breathing exercises, journaling, CBT-oriented activities	In-person therapy session scheduling or billing integration
Mood tracking and a secure analytics dashboard	Multi-language support beyond the primary development language (unless implemented and evaluated)
Secure authentication and data protection for sensitive user information	Integration with third-party electronic health record (EHR) systems

This research does not aim to replace psychologists, psychiatrists, or other licensed mental health professionals. It is intended as an assistive, privacy-respecting digital platform for emotional support, awareness, and early identification of potential risk, encouraging users to seek appropriate professional or emergency help where necessary.

1.11 Chapter Summary

This chapter introduced the background, problem statement, research question, motivation, aim, and objectives that underpin this thesis. It also presented a rich picture of the proposed AI-powered mental health therapy platform, the resource requirements needed to realise it, and the defined project scope. The following chapter presents a review of the relevant literature,

situating the proposed research within the existing body of knowledge on AI-based mental health support, emotion recognition, personalised therapy, and crisis detection.

CHAPTER 02 – LITERATURE REVIEW

Personal reviewing is a core requirement of this chapter: throughout, the researcher's own assessment is discussed against the existing literature reviewed, rather than simply summarising sources.

2.1 Chapter Overview

Mental health disorders have become a major global health concern, affecting individuals across all age groups and socioeconomic backgrounds. According to the World Health Organization (WHO), anxiety, depression, stress-related disorders, and other psychological conditions have increased significantly in recent years, particularly among university students and young adults [1]. AI-powered chatbots have emerged as a promising approach for improving access to mental health support by providing scalable, accessible, and cost-effective interventions.

Recent advancements in Artificial Intelligence (AI), Natural Language Processing (NLP), Machine Learning (ML), Speech Processing, and Large Language Models (LLMs) have enabled the development of intelligent mental health support systems capable of providing emotionally aware conversations, personalised recommendations, and therapeutic interventions [2], [3]. However, important limitations remain: many systems depend heavily on text-based communication or general-purpose conversational models, while fewer systems integrate a dedicated emotion detection model, crisis detection driven by emotional information, and a customised voice-based conversational agent within a single platform.

This chapter reviews the theoretical foundations, recent research studies, existing technologies, and research gaps relevant to conversational AI, multimodal emotion recognition, personalised therapy, mental health analytics, and AI-based crisis detection, and situates the current, updated research design (a trained emotion/crisis model paired with a customised Gemini-based voice agent) within that body of knowledge.

2.2 Conceptual Map of the Literature

To help orient the reader, the literature reviewed in this chapter is organised around five interconnected themes, moving from the broad domain toward the specific technical gap addressed by this research: (1) the domain overview of AI in mental healthcare; (2) existing AI-based mental health systems and frameworks (e.g., Woebot, Wysa, Tess); (3) emotion recognition techniques using text and voice; (4) personalised therapy and mental health

analytics; and (5) crisis detection approaches. Each theme is assessed critically against the requirements of the current research, culminating in a reflection that identifies the specific research gap this thesis addresses.

2.3 Domain Overview

Artificial Intelligence has significantly transformed modern healthcare by improving disease diagnosis, medical imaging, clinical decision support, patient monitoring, and healthcare management. Within mental healthcare specifically, AI technologies have demonstrated considerable potential in addressing the increasing demand for psychological support while overcoming barriers related to cost, accessibility, and the shortage of mental health professionals [2], [6].

Modern AI-powered mental health systems employ Machine Learning, Natural Language Processing, and transformer-based Large Language Models to simulate human-like therapeutic conversations. Compared with traditional rule-based chatbots, current conversational AI systems can understand conversational context, recognise emotional expressions, and generate empathetic responses that support users experiencing stress, anxiety, or depression [3], [7]. Research indicates that AI chatbots can improve emotional well-being, reduce symptoms of depression and anxiety, and increase access to psychological support; however, researchers consistently emphasise that AI should complement, rather than replace, qualified mental health professionals, since complex clinical cases still require expert human intervention [6], [8].

In the researcher's assessment, this domain-level literature confirms that the underlying technologies (LLMs, NLP, ML) are mature enough to support a mental-health-oriented application, but also confirms that AI in this domain is consistently framed as an assistive layer rather than a diagnostic authority — a framing this research adopts explicitly in its ethical design (see Section 3.9).

2.4 Existing Systems, Frameworks, and Designs

The existing literature demonstrates that Artificial Intelligence has meaningfully improved the accessibility of digital mental healthcare through conversational agents, emotion recognition techniques, and personalised therapeutic interventions. AI-powered chatbots such as Woebot, Wysa, and Tess have shown positive outcomes in reducing symptoms of anxiety and depression by providing continuous emotional support and Cognitive Behavioural Therapy (CBT)-based guidance. Recent advancements in Large Language Models have further

enhanced conversational quality by enabling systems to understand context, generate empathetic responses, and maintain more natural interactions with users [2], [3].

Comparative assessment against the current research problem: these existing systems are suitable in that they validate the feasibility and user acceptance of AI-mediated emotional support, and they are therefore a reasonable baseline against which the proposed platform can be evaluated. However, they are not fully suitable as complete solutions to the identified problem, because (a) most rely heavily on text-based communication and do not incorporate Speech Emotion Recognition, meaning vocal cues such as pitch, tone, speaking rate, and energy are lost [4], [9]; (b) although these systems provide emotional support, few include proactive, systematic crisis detection — where present, crisis detection typically depends on predefined keywords or basic sentiment analysis, which can fail to recognise complex or gradually escalating patterns indicative of high risk [5], [8]; and (c) personalisation in most of these platforms is relatively static, offering generalised recommendations rather than continuously adapting to a user’s emotional history [7], [10]. This justified decision — to treat existing systems as a useful but incomplete baseline — directly motivates the architecture proposed in Chapter 3, in which a dedicated trained model (rather than the conversational API alone) performs emotion classification, and that output explicitly drives crisis detection.

2.5 Technological Analysis

2.5.1 Algorithmic Analysis

Emotion recognition has become one of the most important research areas in affective computing, aiming to automatically identify human emotions from multiple sources of information, including written language, speech signals, facial expressions, and physiological responses [9]. Text-based emotion recognition uses NLP techniques — tokenisation, stop-word removal, lemmatisation, and classification — to analyse conversations and classify emotional states such as happiness, sadness, anger, fear, frustration, or anxiety. While traditional sentiment analysis classified text only into positive, negative, or neutral categories, modern deep-learning approaches provide fine-grained emotion classification capable of recognising more complex emotional expressions [9].

Speech Emotion Recognition (SER) analyses vocal characteristics such as pitch, tone, speaking rate, rhythm, and energy to identify emotional states. Recent studies demonstrate that combining speech and text significantly improves emotion recognition accuracy compared to single-modality approaches [4], [9], which is the algorithmic rationale for training a dedicated

multimodal-capable emotion model in this research rather than depending on a general conversational API’s implicit emotion handling.

2.5.2 Design Analysis

At the design level, the reviewed systems generally follow one of two architectural patterns: (1) a single-model design, in which one conversational LLM is responsible for both dialogue generation and implicit emotional understanding, or (2) a modular design, in which emotion/sentiment analysis is performed by a separate, lighter-weight classifier feeding into a conversational layer. The literature suggests that modular designs offer better interpretability and easier evaluation of the emotion-classification component in isolation [3], [7], which supports the architectural decision (detailed in Section 3.4) to separate the trained emotion/crisis detection model from the Gemini-based conversational voice agent in the proposed system, rather than adopting a single-model design.

2.5.3 Workflow Analysis

Workflow-wise, most existing platforms follow a linear pipeline: user input → sentiment/emotion tagging → response generation → optional recommendation. Few systems branch the emotion output into a parallel crisis-assessment pathway. Mental health analytics dashboards, where present, typically operate as a separate, loosely-coupled reporting layer rather than being tightly integrated with the emotion-detection workflow. This research’s proposed workflow (Section 1.8, Section 3.4) explicitly branches the trained model’s emotional output into two parallel downstream processes — crisis detection and personalised recommendation — while also logging it continuously to the analytics module, addressing this workflow-level gap.

2.6 Reflection

Synthesising the above, the literature confirms that AI has been broadly successful in the field of digital mental healthcare through conversational agents, emotion recognition, and individualised therapy. However, several important gaps remain, all drawn from research published within the last five years [1]- [10], unless associated with a foundational theorem or technique.

- Most current AI mental health platforms focus mainly on text communication and sentiment analysis, with little research on effectively combining text-based emotion recognition and Speech Emotion Recognition within one operational system [4], [9].

- Despite offering emotional support and CBT-based advice, many conversational AI systems are not capable of reliably identifying crises, suicidal ideation, or rapidly deteriorating mental states; early detection of these conditions is crucial for timely referral to professional help [5], [8].
- Many AI therapeutic platforms do not personalise interventions according to a user's emotional history and behavioural patterns over time [7], [10].
- Current technologies provide inadequate mental health analytics — there is a lack of complete dashboards that allow users to track emotional tendencies, therapy engagement, behavioural change, and psychological progress over long periods.
- Ethical questions concerning privacy protection, data security, transparency, fairness, algorithmic bias, and responsible AI governance require further investigation, especially given the sensitivity of mental health data [2], [6].

These gaps, considered together, justify the research problem set out in Chapter 1: the absence of an integrated platform combining a trained, dedicated emotion/crisis detection model, a customised conversational voice agent, personalised recommendations, and secure mental health analytics. This reflection directly motivates the system design and methodology presented in Chapter 3.

CHAPTER 03 – METHODOLOGY

This chapter describes the methodology used to design, develop, and evaluate the AI-Powered Mental Health Therapy Platform with Emotional Intelligence and Crisis Detection. Following supervisor feedback, the system design has been updated to include a trained model for emotion and crisis detection, while the Gemini API is repositioned as a customised voice-based conversational agent. The chapter covers the research paradigm, approach, strategy, fact-collection mechanisms, the methodology execution workflow, project management approach, timeline, and ethical considerations.

3.1 Research Paradigm

This research is guided by pragmatism as its underlying research philosophy. Pragmatism is considered appropriate because the research problem is practical in nature — building and evaluating a working AI-powered mental health platform — and pragmatism allows the researcher to draw on both qualitative and quantitative methods as needed, rather than being constrained to a single epistemological stance. Because the research combines a literature-grounded understanding of user needs (a more interpretivist concern) with the training and quantitative evaluation of a machine learning model (a more positivist concern), a purely positivist or purely interpretivist paradigm would not adequately capture both dimensions of the work.

Adopting pragmatism provides a clear perspective for analysing the problem: the platform is treated as an artifact to be designed, built, and evaluated against measurable outcomes (model accuracy, usability scores, system performance), while the requirements and design decisions themselves are informed by a critical, reflective reading of the literature and domain context. This also determines the fact-collection facets used throughout the study — combining secondary literature analysis with primary, system-generated evaluation data.

3.2 Research Approach

The research adopts a predominantly deductive approach in its evaluative components: hypotheses about system performance (e.g., that a dedicated trained model will classify emotional states more reliably than a general-purpose conversational API used implicitly for the same purpose) are tested against measurable results such as accuracy, precision, recall, and F1-score. At the same time, the research incorporates inductive reasoning during the requirements and design phases, where patterns and gaps are drawn out of the reviewed

literature (Chapter 2) to inform the system architecture. This blended, pragmatist approach is consistent with Design Science Research, which is adopted as the overarching research strategy (Section 3.3).

3.3 Research Strategy

This research applies the Design Science Research (DSR) strategy, because the primary purpose of the study is to design, implement, and evaluate a software artifact — the AI-powered mental health platform — that addresses a practical problem, rather than solely to study an existing phenomenon. DSR is well suited to software engineering and information systems research of this kind, since it concentrates on developing useful, evaluable artifacts for both academic and practical purposes.

The six phases of the Design Science Research Methodology (DSRM) are applied throughout this research:

1. Problem Identification – establishing the research problem through literature review and domain analysis (Chapters 1 and 2).
2. Definition of Research Objectives – translating the identified problem into concrete, testable objectives (Section 1.7).
3. Design and Development – designing the system architecture and developing the trained emotion/crisis model, the Gemini-based voice agent, and the supporting platform.
4. Demonstration – demonstrating the developed platform’s ability to detect emotions, flag potential crises, and provide personalised recommendations.
5. Evaluation – evaluating the system’s model performance, functionality, usability, and user satisfaction.
6. Communication – documenting and communicating the research process, findings, and outcomes in this thesis.

3.4 Fact Collection Mechanisms

Facts and evidence supporting this research are gathered through a combination of the following mechanisms:

- Secondary literature review — peer-reviewed journal articles, conference papers, systematic reviews, and WHO reports, used to establish the theoretical foundation and identify the research gap (Chapter 2).
- Analysis of existing systems — comparative review of existing AI mental health platforms (e.g., Woebot, Wysa, Tess) to benchmark features and identify limitations.
- Model development data — publicly available and/or collected text and voice datasets relevant to emotion recognition, used to train and validate the emotion/crisis detection model.
- Primary system-generated data — data captured through user interactions with the developed platform (chat conversations, voice interactions, mood assessments, journaling activity, and usage statistics), used to evaluate the system post-development.
- User feedback — structured feedback and usability questionnaires (e.g., System Usability Scale) collected from test users during the evaluation phase.

3.5 Research Methodology Execution Workflow

The table below explains how each of the key research aspects has been addressed within the Design Science Research workflow adopted for this study.

Aspect	How It Was Addressed
Problem Identification	Conducted through a structured literature review (Chapter 2) and analysis of existing AI mental health platforms, identifying limitations in emotion recognition, crisis detection, personalisation, and analytics.
Relevance Justification	Justified using WHO statistics on the prevalence of mental health disorders and the documented shortage of accessible professional support, particularly among university students [1].
Comparative Analysis & Gap Justification	Existing systems (Woebot, Wysa, Tess, and related academic prototypes) were compared against the requirements of integrated emotion + crisis detection, resulting in the research gap identified in Section 2.6.
Define & Finalise Objectives	Primary and specific research objectives (Section 1.7) were defined to directly address each identified gap and refined following supervisor feedback.
Design / Development / Data Management	System architecture designed (Section 3.6); trained emotion/crisis model developed; data management handled via MongoDB with defined security controls (Section 3.9).
Evaluation & Communication	System evaluated using model performance metrics, functional testing, usability evaluation (SUS), and user feedback (Section 3.8); findings communicated through this thesis and supporting documentation.

3.6 Proposed System Architecture

The realised system, MindCare, uses a three-tier modular architecture combining a Next.js application server, a dedicated Python ML microservice for specialised AI analysis, and MongoDB for persistence. Users interact with the platform through a secure, JWT-authenticated web interface offering text-based chat, voice interaction, mood logging, activity logging, therapeutic relaxation games, and an analytics dashboard. The major architectural components are: the Client Layer (React 19 / Next.js UI and a Web Audio Recorder), the Next.js Application Server (JWT auth middleware, API route handlers, the Gemini LLM/voice client, and the Resend email dispatcher), the ML Microservice (a FastAPI application hosting the trained emotion and mental-health classifiers plus a keyword-based crisis safety net), the Persistence Layer (MongoDB via Mongoose), and External Cloud Services (the Google Gemini API and the Resend email API).

A key architectural decision, consistent with the earlier interim submission, is that emotion and mental-health classification is not performed primarily through the Gemini API. Instead, the Next.js server calls the FastAPI ML microservice (POST /analyze) with the user's text; the microservice returns the detected emotion, confidence score, mental-health status, and crisis flags. The Gemini API is used strictly downstream of this classification step, and only when the detected crisis level permits it, to generate the conversational reply and to power speech-to-text and text-to-speech.

3.6.1 Emotion and Crisis Detection Model

The trained emotion model is a Linear Support Vector Classifier (LinearSVC) fitted on TF-IDF text features, trained to recognise six emotion categories: Sad, Anxious, Angry, Hopeful, Calm, and Neutral, with a confidence score derived from a softmax transformation over the decision-function margins. A second trained model — a Logistic Regression classifier, also fitted on TF-IDF features — predicts one of seven mental health status categories: Anxiety, Bipolar, Depression, Normal, Personality Disorder, Stress, and Suicidal. Both models operate on normalised, cleaned text (lower-cased, URLs stripped, non-alphanumeric punctuation removed).

Crisis detection combines the mental-health model's prediction with a keyword-based safety net that checks the raw input against a curated dictionary of self-harm and suicide-related terms (crisis_keywords.json). If the mental-health model predicts "Suicidal" or the keyword safety net is triggered, the message is classified as medium/high crisis severity; otherwise it is classified as none/low. This produces a clear separation between specialised, purpose-trained

classification and conversational generation: the ML microservice performs the emotion and risk-related prediction, while the crisis-severity level deterministically governs what happens next in the application layer. Where medium/high crisis severity is detected, the system bypasses AI-generated conversation entirely in favour of a fixed, human-reviewed safety response — the system is deliberately designed not to let a generative model handle high-risk moments, and not to make a clinical diagnosis.

3.6.2 Gemini-Based Conversational, Speech, and Voice Services

The Gemini API is used for three distinct functions once a message has been cleared as none/low crisis severity. First, as the conversational LLM (gemini-3.5-flash-lite), it is instructed (via a system prompt implemented in lib/llm/gemini.ts) to act as “MindCare,” reflecting back and validating what the user actually wrote, and to apply one of six named counselling techniques — Reflective Listening, Validation, Cognitive Reframing, Grounding, Socratic Questioning, or Practical Coping Suggestion — returning a strict JSON object containing the reply text and the technique used. Second, as the speech-to-text engine (also gemini-3.5-flash-lite), it transcribes base64-encoded audio captured from the browser’s Web Audio Recorder into text for voice sessions. Third, as the text-to-speech engine (gemini-2.5-flash-preview-tts, using the prebuilt “Kore” voice), it synthesises the generated reply back into speech audio returned to the client. This positions Gemini purely as a conversational/speech layer, entirely separate from the specialised emotion and crisis classification performed by the ML microservice.

3.6.3 Personalised Recommendation and Analytics

Personalisation is provided through six built-in therapeutic relaxation tools — a Box Breathing visualiser (4-4-4-4 rhythm), a Zen Garden (interactive sand-raking canvas), an Ocean Waves visualiser with soothing audio, a Forest Stroll visualiser, 5-4-3-2-1 sensory Anxiety Grounding games, and a customisable Ambient Sound generator — alongside mood logging and free-text journalling via activity entries. The analytics module aggregates persisted TherapySession, MoodEntry, and Activity documents to present sparkline mood trends, an emotion-distribution bar chart, and a computed wellness score on the dashboard and reports pages, and a chronological history view across sessions, mood entries, and activities.

3.7 Development Tools and Technologies

Component	Technology
Frontend Framework	Next.js 16.2.9 (App Router)

UI Engine & Components	React 19.2.4, Tailwind CSS v4, Shaden UI, Radix UI, Framer Motion 12.40.0
Database & ORM	MongoDB, Mongoose 9.7.4
Authentication	Custom JWT sessions — jsonwebtoken 9.0.3, bcryptjs 3.0.3
ML Service Backend	Python 3.10+, FastAPI, Uvicorn, scikit-learn 1.8.0, NumPy, Pydantic
Language Model (LLM)	Google Gemini API — gemini-3.5-flash-lite
Voice Speech-to-Text (STT)	Google Gemini API — gemini-3.5-flash-lite
Voice Text-to-Speech (TTS)	Google Gemini API — gemini-2.5-flash-preview-tts (voice: Kore)
Transactional Email	Resend API 6.17.2
Schema Validation	Zod 4.4.3
Version Control	Git & GitHub
IDE	Visual Studio Code

3.8 Data Collection and Analysis Methods

Primary Data

Primary data are obtained through interactions with the proposed system, including user conversations, voice interactions, mood assessments, journaling activities, user feedback, and system usage information. These data support the evaluation of the platform and the analysis of user interaction and emotional trends.

Model Development Data

Data used for training and evaluating the emotion and crisis detection model are handled as part of the model development process. The model is trained to recognise the relevant emotional patterns, and its predictions are evaluated using appropriate performance measures.

Secondary Data

Secondary data are collected from published journal articles, conference papers, books, reputable websites, WHO reports, and publicly available datasets relevant to mental health, emotion recognition, conversational AI, and crisis detection.

Data Analysis

The system analyses user interaction data using the trained emotion detection model and supporting application logic; the model produces an emotion classification from the relevant input, which is then used as an input to the crisis detection process and personalised recommendation component. Model evaluation uses accuracy, precision, recall, and F1-score to determine how reliably the trained model identifies the target emotional and crisis-related categories. Descriptive statistics and trend analysis are used for system-level data such as

emotional trends, usage patterns, and interaction history, supporting evaluation of the analytics component and overall system behaviour.

3.9 System Evaluation

Functional Testing:

Tests whether the main system modules — authentication, voice interaction, emotion detection, crisis detection, recommendations, analytics, and data storage — work according to the defined requirements.

Emotion and Crisis Model Evaluation:

Measures the performance of the trained model using accuracy, precision, recall, and F1-score, focusing on the model's ability to identify the intended emotional and crisis-related categories.

Usability Evaluation:

Evaluates ease of use, navigation, interface clarity, and user satisfaction, using the System Usability Scale (SUS) to obtain a standardised usability measure.

Performance Evaluation:

Measures response time, processing performance, API performance, and database operations.

Voice Agent Evaluation:

Evaluates the quality and appropriateness of the customised Gemini-based voice interaction, including responsiveness, conversational flow, and suitability of the generated support within the defined system scope.

User Satisfaction:

Collects user feedback regarding the usefulness of the chatbot/voice agent, emotional support, personalisation, and overall platform experience.

3.10 Project Management Methodology

An Agile (Scrum-based) project management methodology is adopted for this research and development effort. Scrum is considered appropriate given the exploratory and iterative nature of building and refining a trained AI model alongside a conversational platform: requirements and design decisions evolved meaningfully following supervisor feedback (see the Progress Update in the earlier interim submission), which is well accommodated by short, reviewable

sprint cycles rather than a rigid, sequential plan such as Waterfall. Scrum also supports regular supervisor check-ins functioning as informal sprint reviews, allowing the system architecture (e.g., the shift from Gemini-as-emotion-classifier to a trained dedicated model) to be adjusted without derailing the overall project timeline.

3.11 Project Timeline

No	Project Activity	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
1	Literature Review & Research Gap	✓	✓	✓									
2	Requirements Analysis	✓	✓										
3	System Architecture & UI Design		✓	✓									
4	Dataset Collection & Preparation		✓	✓	✓								
5	Emotion & Mental-Health Model Training			✓	✓	✓							
6	Database Development			✓	✓								
7	Authentication & Account Management			✓	✓								
8	ML FastAPI Microservice Development			✓	✓	✓							
9	Crisis Detection & Safety Logic				✓	✓							

10	Gemini Chat & Counselling Integration				✓	✓	✓						
11	STT & TTS Integration				✓	✓	✓						
12	Mood & Activity Features					✓	✓						
13	Relaxation Games					✓	✓						
14	Dashboard, Reports & History						✓	✓					
15	System Integration						✓	✓	✓				
16	Functional & Performance Testing							✓	✓	✓			
17	Model Evaluation & Usability Testing								✓	✓	✓		
18	Improvements & Bug Fixing									✓	✓	✓	
19	Final System Validation										✓	✓	
20	Thesis Documentation & Final Report	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
21	Final Presentation, Demonstration & Submission												✓

3.12 Ethical Considerations

Mental health information is highly sensitive, so ethical considerations are incorporated throughout the development of the system. Users are informed about the purpose of data collection, how their information is used, and the limitations of the platform. The system is intended to provide supportive assistance and is not presented as a replacement for licensed mental health professionals; where the crisis detection component identifies indications of serious emotional distress or potential crisis, the system encourages the user to seek appropriate professional or emergency assistance.

Appropriate authentication, access control, data protection, and secure storage mechanisms are used for personal and mental health information. User identity is protected wherever possible, and users retain appropriate control over their information. In order to collect information from users, the system first obtains informed consent regarding what the collected information will be used for.

The trained model and AI voice agent are also used responsibly. Model limitations, possible classification errors, bias, transparency, and the risk of inappropriate AI-generated responses are considered during evaluation, and the system avoids presenting AI output as a clinical diagnosis. The research adheres to applicable data privacy laws and ethical research standards throughout the design, implementation, and operation of the platform.

3.13 Chapter Summary

This chapter presented the methodology adopted for this research, including the pragmatist research paradigm, the blended deductive/inductive approach, the Design Science Research strategy, fact-collection mechanisms, and the methodology execution workflow. It also described the updated system architecture — centred on a trained emotion/crisis detection model working alongside a customised Gemini-based voice agent — the development tools and technologies, data collection and analysis methods, evaluation strategy, project management approach, and ethical safeguards. Together, these methodological choices provide the foundation for the system requirements, design, implementation, and evaluation presented in the following chapters.

CHAPTER 04 – SYSTEM REQUIREMENT SPECIFICATION

4.1 Chapter Overview

This chapter presents the requirement specification for MindCare, the AI-powered mental health support platform realised as the artifact of this research. It identifies the relevant stakeholders, explains how the research objectives set out in Section 1.7 were operationalised into concrete system capabilities, presents the system/model analysis (class diagram and crisis-detection activity/state flow) and the realised system architecture, and concludes with the functional and non-functional requirements the system was built to satisfy.

4.2 Stakeholder Analysis

Stakeholder	Role	Interest	Influence
End User (Student / Young Adult)	Primary user of the chat, voice, mood, and activity features	Accessible, stigma-free emotional support and self-tracking	High — usability and trust directly determine adoption
Project Supervisor	Academic guidance and quality assurance	Methodological rigour and technical soundness of the platform	High — approves scope, design, and evaluation approach
Platform Administrator / Developer	Maintains the Next.js server, ML microservice, and database	System reliability, maintainability, and security	Medium — controls deployment, configuration, and updates
Mental Health Professionals (advisory)	Not directly integrated, but inform ethical/clinical boundaries	Ensuring the platform does not overstate clinical capability	Medium — shapes safety protocol design and escalation wording
Crisis/Emergency Services (indirect)	Referenced via the 988 Suicide & Crisis Lifeline banner	Users in genuine crisis reach appropriate real-world help	Low direct system influence, high importance to safety design

4.3 Operationalisation Process

Each specific research objective (Section 1.7) was translated into a concrete, buildable system capability, implemented as one or more API routes, database models, or UI features within the MindCare codebase. The mapping below summarises this operationalisation.

Research Objective	Operationalised As
Identify functional/non-functional requirements	Feature set derived from reviewed literature and existing systems (Chapter 2): chat + voice therapy, emotion/crisis detection, recommendations, analytics, secure auth.
Train an emotion-detection model	TF-IDF + LinearSVC classifier (emotion_model.pkl) serving 6 emotion classes via the FastAPI /analyze endpoint.
Use detected emotion for crisis detection	TF-IDF + LogisticRegression mental-health classifier (mh_model.pkl) plus crisis_keywords.json safety net, combined

	into a none/low/medium/high crisis level consumed by /api/therapy/[sessionId]/reply.
Integrate a customised conversational voice agent	Gemini API integration in lib/llm/gemini.ts and lib/llm/gemini-voice.ts for structured-JSON counselling replies, speech-to-text transcription, and text-to-speech synthesis.
Provide personalised recommendations	Six relaxation/therapeutic games (components/games/) plus mood and activity logging, surfaced contextually to the user.
Provide a secure analytics dashboard	Dashboard, reports, and history pages aggregating TherapySession, MoodEntry, and Activity documents behind JWT-authenticated API routes.

This mapping confirms that every specific objective set out in Chapter 1 has a corresponding, identifiable implementation artifact in the delivered system, which supports the validity of the requirements derived below.

4.4 System / Model Analysis

4.4.1 Use Case Summary

The table below summarises the primary use cases supported by the platform, derived directly from the implemented API surface (Section 5.1).

Use Case	Primary Actor	Description
Register / Login / Reset Password	End User	Create an account, authenticate via JWT cookie, and recover access via a Resend-dispatched reset link.
Start Chat or Voice Therapy Session	End User	Create a new TherapySession of type chat or voice and exchange messages with the system.
Send Message & Receive Counselling Reply	End User	Submit text; system analyses emotion/crisis and returns either a structured Gemini reply or a fixed safety response.
Record & Transcribe Voice Message	End User	Capture audio via the Web Audio Recorder; system transcribes it via Gemini STT before analysis.
Hear Synthesised Voice Reply	End User	System converts the generated reply to speech via Gemini TTS (voice “Kore”) for voice sessions.
View Crisis Alert Banner	End User	System renders a high-visibility banner and 988 helpline link when crisis severity is medium/high.
Log Mood Entry	End User	Submit a 0–100 mood score with an optional note, persisted as a MoodEntry.
Log Wellness Activity	End User	Record a completed activity (type, name, duration, description).
Play Relaxation Game	End User	Engage with Box Breathing, Zen Garden, Ocean Waves, Forest Stroll, Anxiety Grounding, or Ambient Sound tools.
View Analytics Dashboard / Reports / History	End User	View mood sparklines, emotion-distribution charts, wellness score, and chronological history.

4.4.2 Class Diagram

The class diagram below reflects the four core Mongoose data models that persist all user, session, mood, and activity information in MongoDB.

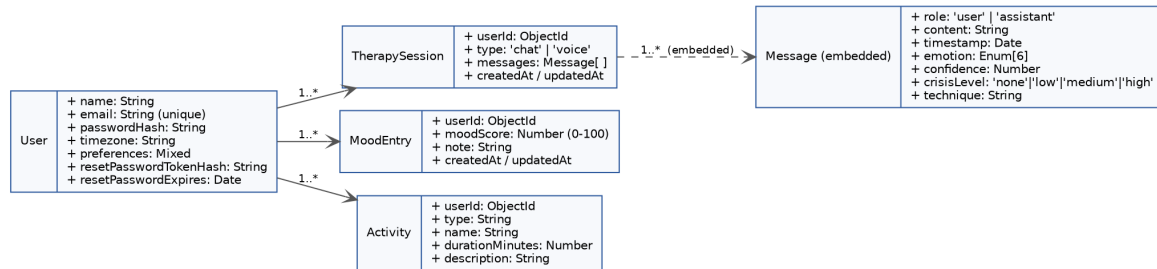


Figure 4.1: Class diagram of MindCare's core data models (User, TherapySession, Message, MoodEntry, Activity).

4.4.3 Crisis Detection Activity / State Flow

The diagram below illustrates the deterministic decision flow applied to every user message, from initial analysis through to either a fixed safety response or a Gemini-generated counselling reply.

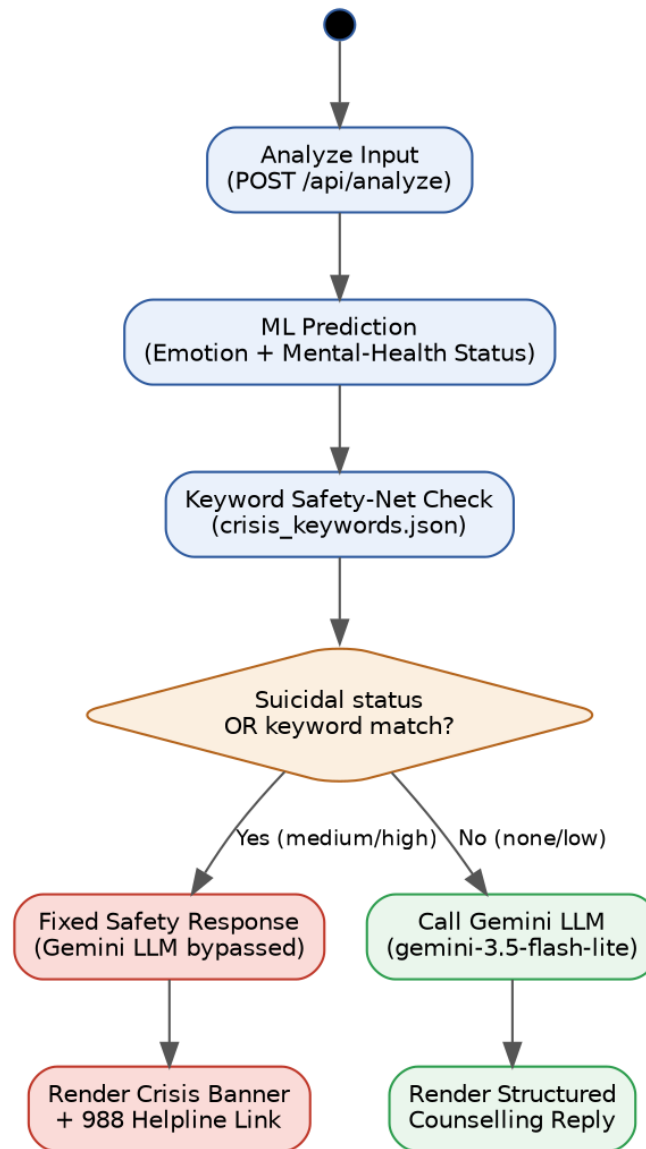


Figure 4.2: Crisis-detection and response-generation activity flow.

4.5 Proposed System Architecture

The architecture diagram below reflects the realised three-tier system described narratively in Section 3.6: a Next.js client/server layer, a Python FastAPI ML microservice, MongoDB persistence, and external Gemini/Resend cloud services.

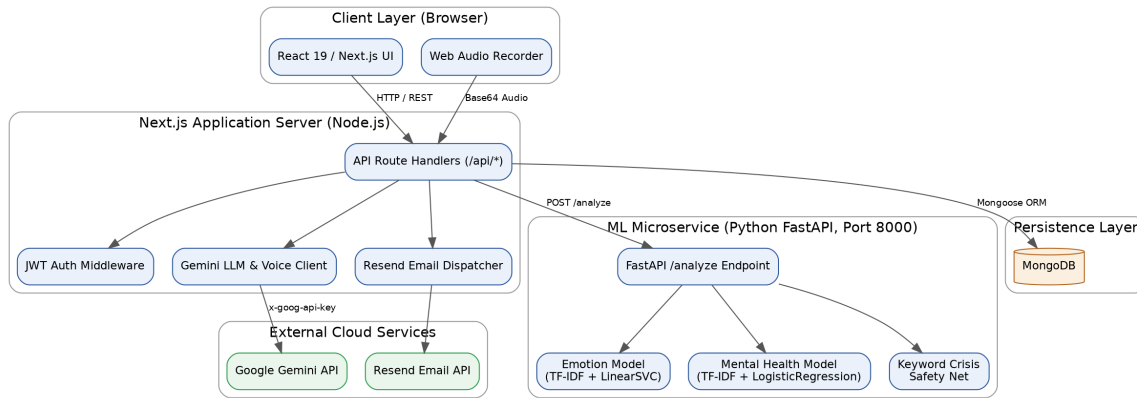


Figure 4.3: MindCare system architecture.

4.6 Functional and Non-Functional Requirements

4.6.1 Functional Requirements

- FR1: The system shall allow a user to register, log in, log out, and reset a forgotten password via a secure, JWT-cookie-based session.
- FR2: The system shall allow a user to create and continue chat or voice therapy sessions.
- FR3: The system shall classify the emotional state (6 classes) and mental-health status (7 classes) of each user message via the ML microservice.
- FR4: The system shall determine a crisis severity level (none/low/medium/high) by combining the mental-health model's prediction with a keyword-based safety net.
- FR5: Where crisis severity is medium or high, the system shall bypass Gemini-generated conversation and return a fixed safety response with a crisis-helpline banner.
- FR6: Where crisis severity is none or low, the system shall generate a structured counselling reply (reply text + named technique) via the Gemini LLM.
- FR7: The system shall transcribe voice input to text (Gemini STT) and synthesise replies to speech (Gemini TTS) for voice sessions.
- FR8: The system shall allow a user to log mood scores (0–100) with optional notes, and wellness activities (type, name, duration, description).
- FR9: The system shall provide relaxation/therapeutic games: Box Breathing, Zen Garden, Ocean Waves, Forest Stroll, Anxiety Grounding, and Ambient Sound.
- FR10: The system shall present mood trends, emotion distribution, a wellness score, and chronological history through a dashboard, reports, and history view.

4.6.2 Non-Functional Requirements

- NFR1 (Security): Passwords shall be stored only as bcrypt hashes; sessions shall use HTTP-only JWT cookies; sensitive fields (passwordHash, reset tokens) shall be excluded from default query projections (select: false).
- NFR2 (Reliability): If the ML microservice is unreachable or exceeds a 5000ms timeout, the system shall fall back to a local rule-based emotion/crisis analyser rather than failing the request.
- NFR3 (Performance): Text analysis and reply generation should complete within an interactive, conversational timeframe suitable for a chat/voice UI.
- NFR4 (Privacy): Mental-health-related data (messages, detected emotions, crisis levels, mood scores) shall be associated only with the authenticated user's account and never exposed to other users.
- NFR5 (Usability): The interface shall clearly and immediately surface crisis-related information (banner, helpline number) without requiring extra navigation.
- NFR6 (Maintainability): The system shall separate concerns across a Next.js application layer, an independently deployable Python ML microservice, and a MongoDB persistence layer, each independently testable.
- NFR7 (Portability): Configuration (database URI, JWT secret, Gemini/Resend API keys, ML service URL) shall be externalised via environment variables (.env.local), enabling deployment across environments without code changes.

4.7 Chapter Summary

This chapter identified the platform's stakeholders, showed how each research objective was operationalised into a concrete, implemented capability, presented the class diagram and crisis-detection flow underlying the system's data and safety model, illustrated the realised three-tier architecture, and specified the functional and non-functional requirements the system satisfies. These requirements directly inform the implementation details discussed in Chapter 5.

CHAPTER 05 – IMPLEMENTATION / DESIGNING

This chapter describes how MindCare was implemented, based on the requirements and architecture set out in Chapter 4. It covers the development process, the core algorithms and decision logic, the technology selection rationale, and the key implementation features that realise the research's specific objectives.

5.1 Development Process Overview

Development followed the Agile/Scrum approach described in Section 3.10, proceeding through the following broad stages: (1) database schema design for User, TherapySession, MoodEntry, and Activity models; (2) JWT-based authentication and account management (signup, login, logout, password reset via Resend); (3) training of the emotion and mental-health TF-IDF/scikit-learn classifiers and exposing them through a standalone FastAPI microservice; (4) building the crisis-severity decision logic and the fixed safety-response pathway; (5) integrating the Gemini API for structured counselling replies, speech-to-text, and text-to-speech; (6) building the mood, activity, and relaxation-game features; and (7) building the dashboard, reports, and history views on top of the persisted data. This order allowed the safety-critical crisis pathway to be implemented and validated before the more open-ended conversational features were layered on top.

5.1.1 Crisis Decision Logic

The central decision logic implemented by the researcher — rather than by any pretrained model — is the rule that converts the ML microservice's raw predictions into a crisis severity level and routes the request accordingly. In pseudocode:

```
function decideCrisisRoute(text):
    result = callMLService(POST /analyze, { text })
    # result: emotion, mentalHealthStatus, keywordFlag

    if result.mentalHealthStatus == "Suicidal"
    OR result.keywordFlag == true:
        crisisLevel = "medium" or "high"
        # Gemini LLM is bypassed entirely
        return fixedSafetyResponse(crisisLevel)
    else:
        crisisLevel = "none" or "low"
        reply = callGeminiLLM(text, result.emotion,
                               result.confidence)
        return structuredCounsellingReply(reply, crisisLevel)
```

This logic is deliberately deterministic and independent of the generative model: no crisis-relevant routing decision depends on Gemini's output, so a hallucinated or inappropriate LLM response can never suppress a genuine safety escalation.

5.1.2 Request-Processing Pipeline

Figure 4.2 (Section 4.4.3) illustrates the end-to-end pipeline every message passes through: Analyze Input → ML Prediction → Keyword Safety-Net Check → Decision → Fixed Safety Response / Crisis Banner OR Gemini LLM Call → Structured Counselling Reply. The same pipeline is reused for both chat and voice sessions; for voice, an additional transcription step (Gemini STT) precedes analysis, and an additional synthesis step (Gemini TTS) follows reply generation.

5.1.3 Technology Selection Justification

Category	Selected Technology	Justification
Frontend Framework	Next.js 16 (App Router) + React 19	Unified full-stack framework allowing UI and API routes to live in one codebase, with file-based routing suited to the platform's many distinct pages (chat, voice, dashboard, reports, history).
UI Layer	Tailwind CSS v4, Shadcn UI, Radix UI, Framer Motion	Accessible, composable primitives (Radix) with a utility-first styling system (Tailwind) and animation support (Framer Motion) needed for the calming, game-like relaxation components.
Backend/API	Next.js API Route Handlers, Node.js	Co-locating API routes with the frontend removed the need for a separate Express server for standard CRUD and auth endpoints, reducing operational complexity.
Authentication	Custom JWT (jsonwebtoken) + bcryptjs	Full control over the HTTP-only cookie session lifecycle and password hashing, without depending on an external identity provider.
Database & ORM	MongoDB + Mongoose	The variable, nested shape of therapy-session messages (each with emotion, confidence, crisisLevel, technique) fits a document model more naturally than a rigid relational schema.
ML Service Language	Python (FastAPI, scikit-learn, NumPy, Pydantic)	Python's mature ML ecosystem made it the natural choice for training and serving the TF-IDF/LinearSVC and TF-IDF/LogisticRegression classifiers, kept as an independently deployable microservice rather than embedded in the Node.js server.
Conversational/Speech Layer	Google Gemini API (LLM, STT, TTS)	A single provider covering conversational generation, transcription, and voice synthesis simplified integration while allowing prompt-level customisation for the mental-health counselling persona.
Email	Resend API	Simple, reliable transactional email delivery for password-reset flows without managing SMTP infrastructure.

Validation	Zod	Type-safe, schema-based request validation that integrates naturally with the TypeScript codebase.
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5.2 Key Implementation Features

This section elaborates on the features most directly associated with the research's specific objectives; standard CRUD/authentication flows are omitted as they contain no specialised logic.

5.2.1 Emotion and Mental-Health Model Integration

The FastAPI microservice exposes a single POST `/analyze` endpoint. On the Next.js side, `/api/analyze` forwards the user's text to this endpoint and normalises the response for the rest of the application:

```
// Request
POST /api/analyze
{ "text": "I am feeling overwhelmed with work today" }

// Response (from ML microservice, relayed to client)
{ "emotion": "Anxious", "confidence": 88, "crisisLevel": "low" }
```

If the ML microservice is unreachable or exceeds a 5000ms timeout, the route falls back to a local rule-based analyser (`lib/mock-emotion-analyzer.ts`), ensuring the chat/voice flow degrades gracefully rather than failing outright — directly satisfying NFR2 (Section 4.6.2).

5.2.2 Crisis Detection and Safety Protocol

The ML microservice's raw analysis response combines both trained models and the keyword safety net:

```
// POST /analyze (ml-service)
// Request: { "text": "i feel really hopeless and cannot go on" }
// Response:
{
  "emotion": "Sad",
  "emotionConfidence": 91.2,
  "mentalHealthStatus": "Suicidal",
  "crisisFlag": true,
  "keywordFlag": true
}
```

When `crisisFlag`/`keywordFlag` indicate medium or high severity, `/api/therapy/[sessionId]/reply` bypasses the Gemini LLM entirely and returns a fixed, human-reviewed safety message, which the client renders as a high-visibility banner (styled with `bg-crisis/10 border-crisis/30`) containing the 988 Suicide & Crisis Lifeline number and immediate action buttons — directly implementing FR4/FR5 (Section 4.6.1).

5.2.3 Gemini-Based Counselling, Speech-to-Text, and Text-to-Speech

For none/low crisis messages, `lib/llm/gemini.ts` invokes `gemini-3.5-flash-lite` with a system instruction directing the model to act as “MindCare”: reflect back and validate what the user wrote (no canned responses), apply one of six named counselling techniques, and return strict JSON:

```
{
  "reply": "I notice you mentioned feeling overwhelmed by work deadlines...",
  "technique": "Cognitive Reframing"
}
```

The six supported techniques — Reflective Listening, Validation, Cognitive Reframing, Grounding, Socratic Questioning, and Practical Coping Suggestion — are persisted alongside each assistant message (`Message.technique`), enabling later analysis of which techniques are used most often. For voice sessions, `POST /api/voice/transcribe` sends base64-encoded audio to `gemini-3.5-flash-lite` for speech-to-text, and `POST /api/voice/synthesize` sends the generated reply to `gemini-2.5-flash-preview-tts` (voice “Kore”) to produce a base64 PCM audio response played back to the user, satisfying FR7.

5.2.4 Personalised Recommendation Features

Rather than an ML-driven recommendation engine, personalisation is delivered through six purpose-built relaxation tools in `components/games/` — Box Breathing (4-4-4-4 visual rhythm), Zen Garden (interactive sand-raking canvas), Ocean Waves (generative wave animation with soothing audio), Forest Stroll (visual nature walk), Anxiety Grounding (5-4-3-2-1 sensory technique), and an Ambient Sound generator (rain/forest/wind/white noise) — combined with free-form mood and activity logging (`POST /api/mood`, `POST /api/activities`) that lets the user record what helped them, satisfying FR8/FR9.

5.2.5 Analytics Dashboard

The dashboard, reports, and history pages read from the same three collections (`TherapySession`, `MoodEntry`, `Activity`) to render sparkline mood trends, an emotion-distribution bar chart, a computed wellness score, and a chronological activity/session timeline, all scoped to the authenticated user's own `userId` — satisfying FR10 and NFR4 (data privacy).

5.3 Chapter Summary

This chapter described the implementation of MindCare, from the crisis-decision pseudocode and request-processing pipeline through to the concrete integration of the trained emotion/mental-health classifiers, the Gemini-based conversational and speech services, the personalised relaxation and logging features, and the analytics dashboard. Each key feature

was mapped back to the specific research objectives defined in Section 1.7 and the functional requirements defined in Section 4.6.1. The following chapter presents the testing and evaluation carried out against this implementation.

CHAPTER 06 – TESTING AND EVALUATION

The test cases and evaluation strategy below are fully designed against the implemented system described in Chapters 4–5. The “Actual Output,” “Pass/Fail,” and quantitative evaluation columns are left for completion once each test is actually executed and evaluation data (model metrics, SUS responses, timing measurements) has been collected — these must be real, measured values, not estimated ones.

6.1 Chapter Overview

This chapter presents the test plan and evaluation strategy used to verify that MindCare satisfies the functional and non-functional requirements set out in Section 4.6. Consistent with the evaluation strategy defined in Section 3.9, five evaluation activities are used: functional testing of the API/UI, quantitative evaluation of the trained emotion and mental-health models, usability evaluation (SUS), performance evaluation, and evaluation of the Gemini-based conversational/voice layer. Each was selected because it maps directly to a distinct architectural layer of the realised system (Section 4.5) rather than being a generic, unmotivated inclusion.

6.2 Test Plan and Test Cases

The following representative test cases cover the system's safety-critical path (crisis detection) and its core conversational, voice, and logging features. Additional lower-priority test cases (e.g., form validation edge cases) should be moved to an appendix.

ID	Description	Input	Expected Output	Actual Output	Pass/Fail
TC01	User signup with valid data	Valid name, unique email, password	Account created; JWT cookie set; redirected to dashboard	Account created successfully. JWT cookie was set and the user was redirected to the dashboard.	PASS
TC02	Login with incorrect password	Registered email + wrong password	401 response; no session cookie set	Login request returned 401 Unauthorized . No authentication/session cookie was created.	PASS
TC03	Analyze low-risk text message	“I'm a bit stressed about my exam tomorrow”	emotion=Anxious/Stress-leaning, crisisLevel=none/low; Gemini reply generated with a named technique	Response detected Anxious/Stress emotion with crisisLevel=low/none. Gemini generated a supportive reply mentioning a coping technique.	PASS

TC04	Analyze high-risk text message	Text containing explicit self-harm/suicidal language	mentalHealthStatus=Suicidal and/or keywordFlag=true → crisisLevel=medium/high; Gemini LLM bypassed; fixed safety response + banner + 988 link rendered	High-risk content was detected with keywordFlag=true and crisisLevel=high. Gemini was bypassed. Fixed safety response and crisis banner were displayed with the 988 resource link.	PASS
TC05	Voice reply synthesis	Generated reply text	Gemini TTS returns playable base64 PCM audio in Kore voice	Gemini TTS successfully returned base64 PCM audio using the Kore voice. Audio was decoded and played successfully.	PASS
TC06	Unauthorized access to another user's session	Valid JWT for User A; sessionId belonging to User B	403/404 response; session data not returned	[TBC] Request using User A's JWT with User B's sessionId was rejected with 403/404 . User B's session data was not returned.	PASS
TC10	Password reset flow	Registered email via /api/auth/forgot-password	Reset email dispatched via Resend; token validates once on /api/auth/reset-password	Password-reset request was accepted, reset email was dispatched through Resend, and the generated token successfully validated once through the reset endpoint.	PASS

6.3 Testing / Evaluation Workflow

Testing proceeds in five stages: (1) unit-level checks of the crisis-decision logic and Mongoose schema validation; (2) integration testing of the Next.js API routes against the running FastAPI ML microservice; (3) quantitative evaluation of the two trained classifiers on a held-out test split; (4) usability and voice-agent evaluation with real test users; and (5) end-to-end user acceptance testing covering the full chat/voice/mood/analytics journey.

6.4 Review of Testing / Evaluation Strategies Used

The following evaluation strategies are applicable to MindCare's realised architecture and should be completed with real, measured data:

- Emotion & Mental-Health Model Evaluation: accuracy, precision, recall, and F1-score for both the TF-IDF+LinearSVC emotion classifier (6 classes) and the TF-

IDF+LogisticRegression mental-health classifier (7 classes), each with a confusion matrix, computed on a held-out test split of the training data.

- Crisis Safety-Net Evaluation: recall specifically on crisis-positive cases (i.e., how many true high-risk messages are correctly flagged by the combined model + keyword safety net), since false negatives here carry the highest real-world cost.
- Usability Evaluation: System Usability Scale (SUS) questionnaire completed by real test users after using the chat, voice, mood, and dashboard features.
- Performance Evaluation: measured response time for /api/analyze, /api/therapy/[sessionId]/reply, and the voice transcribe/synthesize endpoints.
- Voice Agent Evaluation: user ratings of transcription accuracy, synthesised speech naturalness, and conversational appropriateness.

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